

Code-Layout, Readability, and Reusability Assessment

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Code Layout Checklist

S. No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing and tabs used throughout frontend JSX and backend Python files.	Yes	Complies with PEP 8 and ESLint configurations.
2	Proper File Structure	Folders logically organized into app, routers, models, schemas, pages, and components.	Yes	Decoupled layout folders.
3	Meaningful Variable Names	Variables and parameters reflect their mathematical or model purpose clearly.	Yes	Self-documenting names.
4	Function / Method Names	Functions are descriptively named using action verbs (e.g. build_pdf, check_user).	Yes	Clean verbs.
5	Code Comments	Inline explanations and block comments explain logical paths and algorithms.	Yes	Preserved code explanations.
6	Modular Design	Code is split into thin routers, custom services, and reuse components.	Yes	Follows separation of concerns.
7	No Redundant Code	Duplicate queries and CSS codes are eliminated via utilities.	Yes	Shared styles classes.
8	Error Handling	Exceptions are caught using try-except, showing clear UI error notifications.	Yes	Graceful try-catches.

Reusable Components / Modules

S. No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Med/Low)
1	Sidebar Layout Drawer	JavaScript (React)	Guarded routes screens layout shell	High
2	ProtectedRoute Guard	JavaScript (React)	Binds all private dashboard routes	High
3	Custom Axios API Client	JavaScript (Axios)	All backend API data communications	High
4	PBKDF2 Password Encrypter	Python (FastAPI)	Auth routers for login & registration	High
5	Gemini AI Query Service	Python (FastAPI)	Advisor chat and recommend routes	High
6	ReportLab Custom PDF builder	Python (ReportLab)	Dossier report & template generators	High